

- Decision Tree is a Supervised Machine Learning algorithm used for classification & regression problems.
- It works by splitting data into smaller subsets based on different conditions.
- The structure looks like an inverted tree, starting from a root node & splitting into branches & leaves.

Concept

- Each internal node represents a test on an attribute (a question)
- Each branch represents the outcome of the test.
- Each leaf node represents a final decision or class label.

Eg. - Question in a decision tree:
"Is the temperature hot, mid, or cool?"

Depending on the answer, the tree moves to the next question until it reaches a decision (like play, don't play).

Numerical

Decision Tree (ID3 Algorithm)

Numerical Example

Day	Outlook	Temp	Humidity	Wind	Target
					Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$N = 14$ (Total no. of data)

Play/Ten.

Yes & No

(Calculate

Entropy of whole dataset, $N = 14$

Yes $\rightarrow 9$
No $\rightarrow 5$

$$\text{Entropy}(S) = - \sum_{i=1}^n P_i \log_2(P_i)$$

$$= - \left(\frac{9}{14} \log_2 \left(\frac{9}{14} \right) + \frac{5}{14} \log_2 \left(\frac{5}{14} \right) \right)$$

$$= 0.94$$

Now, 2) Calculate Entropy for each feature (attribute)
 but node will have highest gain

Attribute: Outlook

Values (Outlook) = Sunny, Overcast, Rain

Entropy(S_{sunny}) =

	Play	Yes	No
outlook: sunny → 5		2	3
overcast → 4		4	0
rain → 5		3	2

$$\text{Entropy}(S_{\text{sunny}}) = - \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right)$$

$$= 0.971$$

$$\text{Entropy}(S_{\text{overcast}}) = - \left(\frac{4}{4} \log_2 \left(\frac{4}{4} \right) + \frac{0}{4} \log_2 \left(\frac{0}{4} \right) \right)$$

$$= 0$$

(pure)

$$\text{Entropy}(S_{\text{rain}}) = - \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.971$$

Now, IG($S, \text{attribute}$)

$$\text{Gain}(S, \text{Outlook}) = \text{Entropy}(S) - \sum_{v \in \{\text{sunny, overcast, rain}\}} \left(\frac{|S_v|}{|S|} \text{Entropy}(S_v) \right)$$

$$= \text{Entropy}(S) - \left(\frac{5}{14} \text{Entropy}(S_{\text{sunny}}) + \frac{4}{14} \text{Entropy}(S_{\text{overcast}}) + \frac{5}{14} \text{Entropy}(S_{\text{rain}}) \right)$$

$$= 0.94 - \left(\frac{5}{14} * 0.971 + \frac{4}{14} * 0 + \frac{5}{14} * 0.971 \right)$$

$$\text{Gain}(S, \text{outlook}) = 0.94 - 0.694$$

$$= 0.246$$

Attribute: Temp

Values(Temp) = Hot, Mild, Cool

Temp:		yes	no
hot	4	2	2
mild	6	4	2
cool	4	3	1

$$\text{Entropy}(S_{\text{hot}}) = - \left(\frac{2}{4} \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right)$$

= 1 (Directly, when classes
are equal)

$$\text{Entropy}(S_{\text{mild}}) = - \left(\frac{4}{6} \log_2 \left(\frac{4}{6} \right) + \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right)$$

$$= 0.918$$

$$\text{Entropy}(S_{\text{cool}}) = - \left(\frac{3}{4} \log_2 \left(\frac{3}{4} \right) + \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right)$$

$$= 0.811$$

Information Gain

Entropy(S)

Entropy(S_i)

$$\text{Gain}(S, \text{Temp}) = H(S) - \sum_{i \in \{\text{hot, mild, cool}\}} \frac{|S_i|}{|S|} H(S_i)$$

$$= 0.94 - \left(\frac{4}{14} * 1 + \frac{6}{14} * 0.918 + \frac{4}{14} * 0.811 \right)$$

$$= 0.94 - 0.911$$
$$= 0.029$$

Again,

Attribute: humidity

value (humidity): ~~weak, strong~~
high, normal

humidity		Yes	No
→	high	7	4
	normal	7	1

$$\text{Entropy}(\text{high}) = - \left(\frac{3}{7} \log_2 \left(\frac{3}{7} \right) + \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right)$$

$$= 0.985$$

$$\text{Entropy}(S_{\text{normal}}) = - \left(\frac{6}{7} \log_2 \left(\frac{6}{7} \right) + \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right)$$

$$= 0.592$$

Now,

$$G_{\text{an}}(S, \text{humidity}) = H(S) - \sum_{v \in \{\text{high, normal}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.94 - \left(\frac{7}{14} \times 0.985 + \frac{7}{14} \times 0.592 \right)$$

$$= 0.152$$

Attribute: wind

value (wind): weak, strong

wind		Yes	No
→	weak	8	2
	strong	6	3

Now,

$$\text{Entropy}(S_{\text{weak}}) = - \left(\frac{6}{8} \log_2 \left(\frac{6}{8} \right) + \frac{2}{8} \log_2 \left(\frac{2}{8} \right) \right)$$

$$= 0.811$$

$$\text{Entropy}(S_{\text{strong}}) = - \left(\frac{8}{6} \log_2 \left(\frac{8}{6} \right) + \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right)$$

$$= 1 \quad (\text{Directly})$$

(both equal)

Now,

$$G_{\text{an}}(S, \text{wind}) = H(S) - \sum_{v \in \{\text{weak, strong}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.94 - \left(\frac{8}{14} \times 0.811 + \frac{7}{14} \times 1 \right)$$

$$= 0.94 - 0.906$$

$$= 0.035$$

Highest Information Gain is,

of outlook $\rightarrow 0.2464$

So, we consider outlook a root node.

S D1, D2, D4, D5, D8, D9, D11

[9 $\rightarrow$ Yes]
[5 $\rightarrow$ No]

Outlook

Sunny

[D1, D2, D8, D9, D11]

5

2 $\rightarrow$ Yes

3 $\rightarrow$ No

1
2
?

Overcast

[D3, D4, D12, D13]

4

4 $\rightarrow$ Yes

0 $\rightarrow$ No

Yes

Rain

[D6, D7, D10, D14]

5

3 $\rightarrow$ Yes

2 $\rightarrow$ No

?

For sunny only dataset.

Recursively Repeat the process

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

(Remember, there is no outlook column here)

Entropy of this dataset,

$$\text{Entropy}(S_{\text{sunny}}) = - \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right)$$

$$= 0.971$$

Now,

Attribute: - Temp

	Yes	No
Temp $\rightarrow$ Hot	2	0
Mild	2	1
Cool	1	1

values $\rightarrow$ Hot, mild, cool

Entropy(S_{Hot}) = Directly 0 (one sided)

Entropy(S_{Mild}) = Directly 1 (Both equal)

Entropy(S_{Cool}) = Directly 0 (one sided)

$$Gain(S_{\text{Sunny}}, \text{Temp}) = H(S_{\text{Sunny}}) - \sum_{v \in \{\text{hot, mild, cool}\}} \frac{|S_v|}{|S_{\text{Sunny}}|} * H(S_v)$$

$$= 0.971 - \left(\frac{2}{5} * 0 + \frac{2}{5} * 1 + \frac{1}{5} * 0 \right)$$

$$= 0.571$$

Again,

for Attribute Humidity,

values(humidity) $\rightarrow$ High, Normal

humidity $\rightarrow$		Yes	No
High	3	0	3
Normal	2	2	0

Entropy(High) = Directly 0 (one sided)

Entropy(Normal) = Directly 1 (one sided)

Gain($S_{\text{Sunny}}, \text{Humidity}$)

$$= H(S_{\text{Sunny}}) - \sum_{v \in \{\text{high, normal}\}} \frac{|S_v|}{|S_{\text{Sunny}}|} * H(S_v)$$

$$= 0.971 - \left(\frac{3}{5} * 0 + \frac{2}{5} * 0 \right)$$

$$= 0.971$$

Again,

for Attribute Wind,

values(wind) $\rightarrow$ weak, strong

		Yes	No
Wind $\rightarrow$ weak	3	1	2
strong	2	1	1

$$\text{Entropy(weak)} = - \left(\frac{1}{5} \log_2 \left(\frac{1}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.918$$

Entropy(strong) = directly 1 (both equal)

$$G_{\text{an}}(S_{\text{sunny}}, \text{wind}) = H(S_{\text{sunny}}) - \sum_{v \in \{\text{strong, weak}\}} \frac{|S_v|}{|S_{\text{sunny}}|} \times \frac{W(S_v)}{W(S_{\text{sunny}})}$$

$$= 0.971 - \left(\frac{3}{5} \times 0.918 + \frac{2}{5} \times 1 \right)$$

$$= 0.971 - 0.951$$

$$= 0.020$$

~~Any~~

$$G_{\text{an}}(S_{\text{sunny}}, \text{Temp}) = 0.570$$

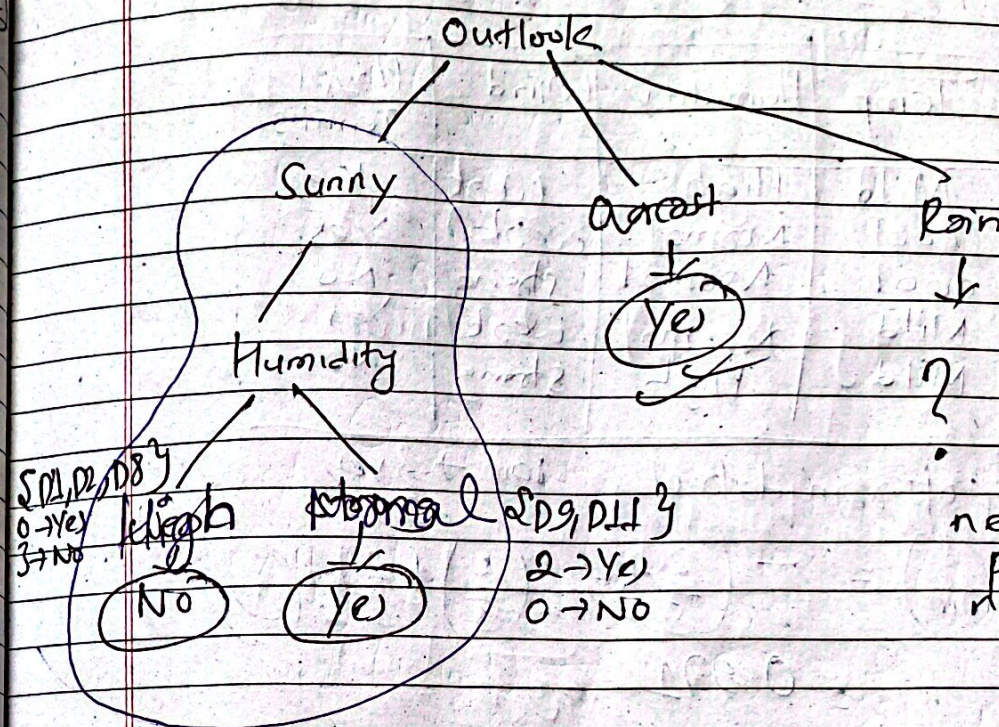
Info.

$$G_{\text{an}}(S_{\text{sunny}}, \text{Humidity}) = 0.97 \rightarrow \text{Highest } G_{\text{an}}$$

So, next node.

$$G_{\text{an}}(S_{\text{sunny}}, \text{Wind}) = 0.0192$$

Humidity	Play tennis
High	No
High	No
High	No
Normal	Yes
Normal	Yes



next part, next page.

Directly
(observed)

Now, for Ran node
Rain only dataset

Date _____
 Page _____

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	weak	Yes
D5	cool	Normal	weak	Yes
D6	cool	Normal	strong	No
D10	Mild	Normal	weak	Yes
D14	Mild	High	strong	No

Entropy of rain dataset rain → 3 yes, 2 no

$$H(S_{\text{rain}}) = - \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.971$$

Again,

for attribute temp,

values(temp) = mild, cool, hot

temp	hot	0	Yes	No
	mild	3	2	1
	cool	2	1	1

Date _____
 Page _____

$$S_{\text{Entropy}}(S_{\text{temp}}) = 0$$

$$\text{Entropy}(S_{\text{mild}}) = - \left(\frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right)$$

$$= 0.918$$

Again,

$$\text{Entropy}(S_{\text{cool}}) = 1 \text{ directly (both equal)}$$

Again,

$$H(S_{\text{rain, temp}}) = H(S_{\text{rain}}) - \sum_{v \in \{\text{rain, cool, hot}\}} \frac{|S_v|}{|S_{\text{rain}}|} \times H(S_v)$$

$$= 0.971 - \left(0 \times 0 + \frac{3}{5} \times 0.918 + \frac{2}{5} \times 1 \right)$$

$$= 0.020$$

Again,

for attribute Humidity,

values(Humidity) = Normal, High

Humidity	Normal	3	Yes	No
	High	2	1	1

$$So, \text{Entropy}(S_{\text{normal}}) = - \left(\frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right)$$

$$= 0.918$$

$$\text{Entropy}(S_{\text{high}}) = 1 \text{ Directly (Both equal)}$$

$$\text{Now, } G_{\text{an}}(S_{\text{ran}}, \text{humidity}) = H(S_{\text{ran}}) - \sum_{v \in \{\text{normal, high}\}} \frac{|S_v|}{|S_{\text{ran}}|} \times H(S_v)$$

$$= 0.971 - \left(\frac{3}{5} \times 0.918 + \frac{2}{5} \times 1 \right)$$

$$= 0.020$$

Again,

$$So, G_{\text{an}}(S_{\text{ran}}, \text{Temp}) = 0.020$$

$$G_{\text{an}}(S_{\text{ran}}, \text{humidity}) = 0.020$$

$$G_{\text{an}}(S_{\text{ran}}, \text{wind}) = 0.971 \quad \text{If best so far next node}$$

for

Attribute wind,

Values (wind) $\rightarrow$ weak, strong
Yes no

wind $\rightarrow$ weak	3	3	0
strong	2	0	2

$$\text{Entropy}(S_{\text{weak}}) = - \left(\frac{3}{3} \log_2 \left(\frac{3}{3} \right) \right) \text{ directly (one sided)}$$

$$\text{Entropy}(S_{\text{strong}}) = 0 \text{ directly (one sided)}$$

$$\text{Now, } G_{\text{an}}(S_{\text{ran}}, \text{wind}) = H(S_{\text{ran}}) - \sum_{v \in \{\text{weak, strong}\}} \frac{|S_v|}{|S_{\text{ran}}|} \times H(S_v)$$

$$= H(S_{\text{ran}})$$

$$= 0.971$$

$\{P_1, P_2, \dots, P_{14}\}$ $yes \rightarrow 9$
 $no \rightarrow 5$

Date _____
Page _____

outlook

Sunny

Overcast

Rainy

Humidity

Yes

Wind

High

Normal

Weak

Strong

NO

Yes

Yes

no

$\{P_3, P_7, P_8\}$

$\{P_9, P_{11}\}$

$\{P_4, P_5, P_{10}\}$

$\{P_6, P_{14}\}$

✓